1D CNN — Architecture Variant Comparison

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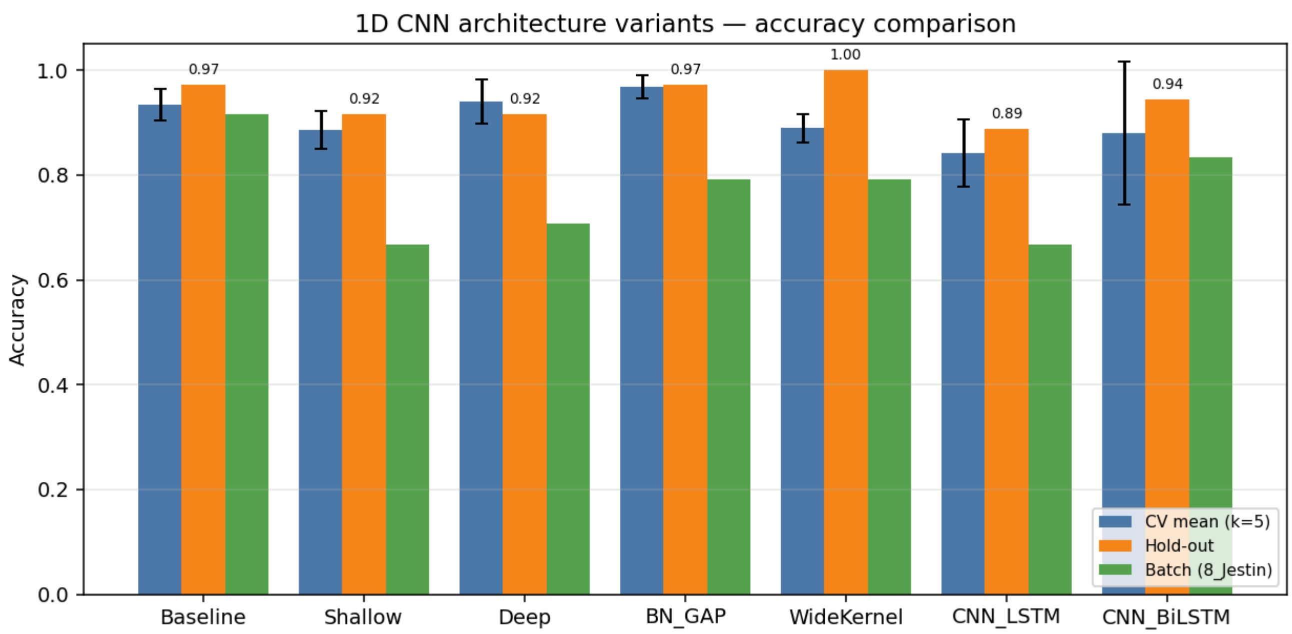
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Total trials	352 (train pool 316, hold-out 36)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 15.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

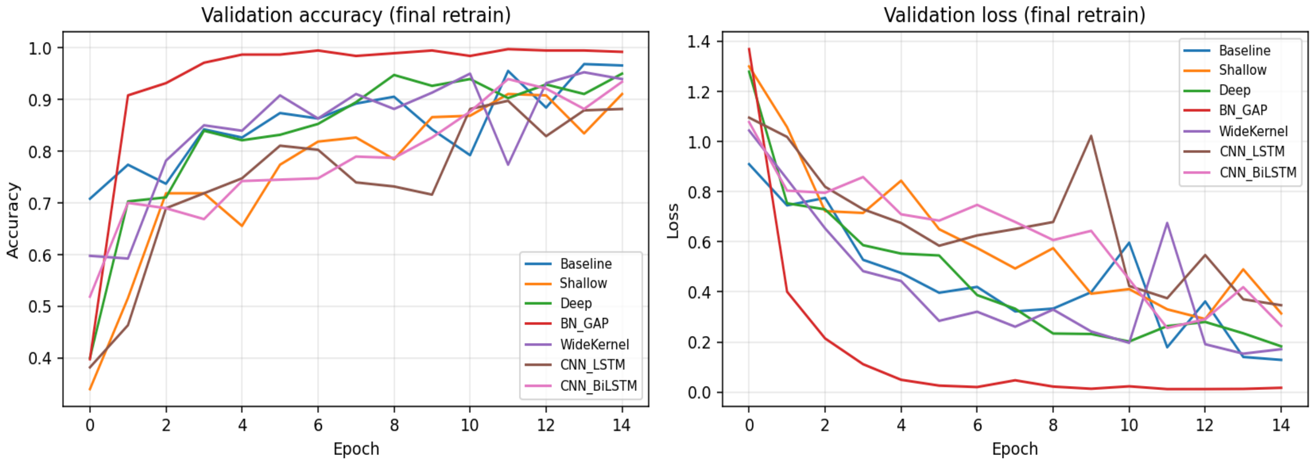
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9337	0.0303	0.9722	0.0579	0.9167	22/24
Shallow	72,487	0.8861	0.0366	0.9167	0.3260	0.6667	16/24
Deep	204,007	0.9401	0.0425	0.9167	0.1257	0.7083	17/24
BN_GAP	57,319	0.9684	0.0223	0.9722	0.0873	0.7917	19/24
WideKernel	131,687	0.8892	0.0266	1.0000	0.0500	0.7917	19/24
CNN_LSTM	68,455	0.8421	0.0647	0.8889	0.3187	0.6667	16/24
CNN_BiLSTM	105,575	0.8806	0.1361	0.9444	0.1550	0.8333	20/24

Best on hold-out: **WideKernel** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Validation accuracy (final retrain)
Accuracy
Epoch
Baseline
Shallow
Deep
BN_GAP
WideKernel
CNN_LSTM
CNN_BiLSTM
Validation loss (final retrain)
Loss
Epoch

Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9337 ± 0.0303
Hold-out test acc / loss	0.9722 · 0.0579
Batch-test acc	0.9167 (22/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1512	64	0.8906	0.9219	0.8906
2	1518	63	0.9524	0.9683	0.9524
3	1518	63	0.9048	0.9365	0.9048
4	1518	63	0.9524	0.9524	0.9524
5	1518	63	0.9683	0.9683	0.9683

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*0.976*	*0.971*	*0.971*	*36*
weighted avg	*0.977*	*0.972*	*0.972*	*36*

Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	4	5	0.800
Double_Wiggle	4	4	1.000
Double_cmere	4	5	0.800
Overall	**22**	**24**	**0.917**

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8861 ± 0.0366
Hold-out test acc / loss	0.9167 · 0.3260
Batch-test acc	0.6667 (16/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1512	64	0.8750	0.8750	0.8750
2	1518	63	0.9206	0.9683	0.9206
3	1518	63	0.8413	0.8889	0.8413
4	1518	63	0.8571	0.9206	0.8571
5	1518	63	0.9365	0.9365	0.9365

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.833	1.000	0.909	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.714	1.000	0.833	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	0.600	0.750	5
macro avg	*0.935*	*0.914*	*0.912*	*36*
weighted avg	*0.937*	*0.917*	*0.914*	*36*

Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	0	5	0.000
Double_Wiggle	4	4	1.000
Double_cmere	2	5	0.400
Overall	**16**	**24**	**0.667**

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9401 ± 0.0425
Hold-out test acc / loss	0.9167 · 0.1257
Batch-test acc	0.7083 (17/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1512	64	0.8750	0.9375	0.8750
2	1518	63	0.9683	1.0000	0.9683
3	1518	63	0.9048	0.9365	0.9048
4	1518	63	0.9683	0.9841	0.9683
5	1518	63	0.9841	0.9841	0.9841

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.833	1.000	0.909	5
Double_Nothing	0.750	0.600	0.667	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.800	0.800	0.800	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*0.912*	*0.914*	*0.911*	*36*
weighted avg	*0.914*	*0.917*	*0.913*	*36*

Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	0	5	0.000
Double_Wiggle	4	4	1.000
Double_cmere	3	5	0.600
Overall	**17**	**24**	**0.708**

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9684 ± 0.0223
Hold-out test acc / loss	0.9722 · 0.0873
Batch-test acc	0.7917 (19/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1512	64	0.9531	0.9531	0.9531
2	1518	63	0.9683	0.9841	0.9683
3	1518	63	0.9365	0.9683	0.9365
4	1518	63	1.0000	1.0000	1.0000
5	1518	63	0.9841	1.0000	0.9841

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*0.976*	*0.971*	*0.971*	*36*
weighted avg	*0.977*	*0.972*	*0.972*	*36*

Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	0	5	0.000
Double_Wiggle	4	4	1.000
Double_cmere	5	5	1.000
Overall	**19**	**24**	**0.792**

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.8892 ± 0.0266
Hold-out test acc / loss	1.0000 · 0.0500
Batch-test acc	0.7917 (19/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1512	64	0.8906	0.9062	0.8906
2	1518	63	0.9206	0.9841	0.9206
3	1518	63	0.8889	0.9365	0.8889
4	1518	63	0.8413	0.9048	0.8413
5	1518	63	0.9048	0.9206	0.9048

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*1.000*	*1.000*	*1.000*	*36*
weighted avg	*1.000*	*1.000*	*1.000*	*36*

Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_Lower	2	5	0.400
Double_Pistol_Recoil	5	5	1.000
Double_Raise	3	5	0.600
Double_Wiggle	4	4	1.000
Double_cmere	5	5	1.000
Overall	**19**	**24**	**0.792**

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,455
CV mean ± std	0.8421 ± 0.0647
Hold-out test acc / loss	0.8889 · 0.3187
Batch-test acc	0.6667 (16/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1512	64	0.7500	0.9062	0.7500
2	1518	63	0.7937	0.9206	0.7937
3	1518	63	0.8730	0.9048	0.8730
4	1518	63	0.8571	0.9048	0.8571
5	1518	63	0.9365	0.9365	0.9365

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.556	1.000	0.714	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	0.600	0.750	5
macro avg	*0.937*	*0.886*	*0.888*	*36*
weighted avg	*0.938*	*0.889*	*0.891*	*36*

Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	1	5	0.200
Double_Wiggle	4	4	1.000
Double_cmere	1	5	0.200
Overall	**16**	**24**	**0.667**

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,575
CV mean ± std	0.8806 ± 0.1361
Hold-out test acc / loss	0.9444 · 0.1550
Batch-test acc	0.8333 (20/24)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1512	64	0.6094	0.9375	0.6094
2	1518	63	0.9365	0.9365	0.9365
3	1518	63	0.9365	0.9365	0.9365
4	1518	63	0.9683	0.9683	0.9683
5	1518	63	0.9524	0.9683	0.9524

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.833	1.000	0.909	5
Double_Nothing	1.000	0.600	0.750	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	*0.952*	*0.943*	*0.938*	*36*
weighted avg	*0.954*	*0.944*	*0.940*	*36*

Batch test per class (8_Jestin)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	4	5	0.800
Double_Wiggle	4	4	1.000
Double_cmere	2	5	0.400
Overall	**20**	**24**	**0.833**